Submission – Group 15

Project Name: Hospital Management System (HMS)
Group Name: Group 15
Course: Scalable Services – Microservices-Based Application Development

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1. Group Details & Individual Contribution

S.No	Name	Email	Contribution (%)	Key Responsibilities
1	ADITYA SEKHAR	2025tm93080@wilp.bits-pilani.ac.in	20%	Patient Service & Appointment Service development; Minikube cluster setup; Kubernetes manifests; deployment orchestration
2	MANASA BHAT	2025tm93082@wilp.bits-pilani.ac.in	20%	Billing Service development; Prometheus & Grafana monitoring stack; metrics endpoint implementation
3	SHARMA VANDANA DINESHCHANDRA SHARDA	2025tm93207@wilp.bits-pilani.ac.in	20%	Prescription Service development; Docker Compose infrastructure; inter-service communication testing
4	PRASANNA VENKAT	2024tm93611@wilp.bits-pilani.ac.in	20%	Doctor Schedule Service development; CSV dataset preparation; seed script (`seed_all.py`); data normalization
5	UPPU VINOD KUMAR	2025tm93233@wilp.bits-pilani.ac.in	20%	Demo UI (React) development; Swagger UI integration; end-to-end workflow testing; documentation

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2. Brief Application Description

The **Hospital Management System (HMS)** is a microservices-based application that manages core hospital operations. It is built with Python/Flask, containerized with Docker, and orchestrated on Kubernetes (Minikube).

What the system does:

- **Patient Service** – Registers patients and manages patient profile data (name, email, phone, DOB).
- **Doctor Schedule Service** – Manages doctor profiles, specializations, and availability slots.

- **Appointment Service** – Creates, updates, reschedules, and tracks patient appointments with doctors.
- **Prescription Service** – Issues and manages prescriptions linked to completed appointments.
- **Billing Service** – Creates bills for appointments and records payments against bills.
- **Demo UI** – A React-based console that demonstrates all service capabilities, validation rules, workflow evidence, and live API responses.
- **Monitoring** – Prometheus scrapes `/metrics` endpoints from all five services; Grafana visualizes dashboards.

Key architectural decisions:

- **Database-per-service** pattern: each microservice owns its own SQLite database with a dedicated PersistentVolume in Kubernetes.
- **Synchronous HTTP/REST** inter-service communication: services validate dependent resources through REST calls rather than sharing databases (e.g., Appointment Service checks Patient Service and Doctor Schedule Service before booking).
- **Containerized & Orchestrated**: each service has its own Dockerfile; the entire stack runs on Minikube with Kubernetes Deployments, Services (ClusterIP), ConfigMaps, Secrets, and PersistentVolumeClaims.

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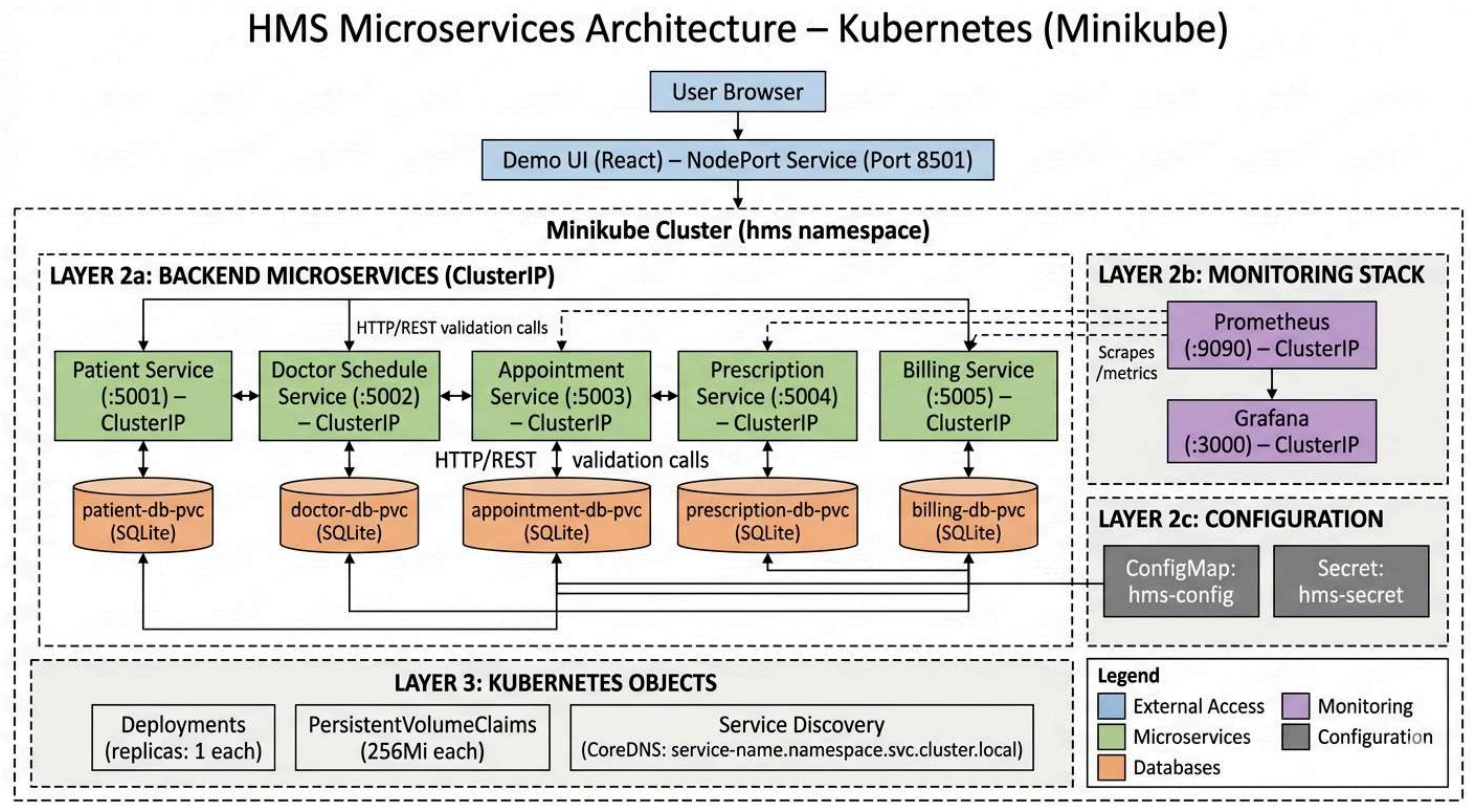
3. System Architecture

3.1 Architecture Diagram



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4. Step-by-Step Execution Instructions

Prerequisites

- **Docker Desktop** installed and running
- **Minikube** installed (`winget install Kubernetes.minikube`)

- **kubectl** installed and available on PATH
- **Git** installed
- **Python 3.11+** installed

4.1 Clone the Consolidated Repository

```
git clone https://github.com/adi-bits-2025/ss-assignment-3.git
cd ss-assignment-3
```

4.2 Start Minikube Cluster

```
# Clean any previous instance
minikube delete --all --purge
docker rm -f minikube 2>$null

# Pre-pull the base image to avoid network hangs
docker pull gcr.io/k8s-minikube/kicbase:v0.0.50

# Start Minikube with Docker driver
minikube start --driver=docker --base-image="gcr.io/k8s-minikube/kicbase:v0.0.50"
```

Verify the cluster:

```
kubectl cluster-info
kubectl get nodes
```

```
C:\Users\HP\Documents\WILP\Sem II\ScalableServices\Assignment\ss-assignment-3>kubectl cluster-info
Kubernetes control plane is running at https://127.0.0.1:60327
CoreDNS is running at https://127.0.0.1:60327/api/v1/namespaces/kube-system/services/kube-dns:dns/proxy

To further debug and diagnose cluster problems, use 'kubectl cluster-info dump'.

C:\Users\HP\Documents\WILP\Sem II\ScalableServices\Assignment\ss-assignment-3>kubectl get nodes
NAME       STATUS   ROLES           AGE     VERSION
minikube   Ready    control-plane   3h34m   v1.35.1

C:\Users\HP\Documents\WILP\Sem II\ScalableServices\Assignment\ss-assignment-3>
```

4.3 Build Docker Images Inside Minikube

Point your shell to Minikube's Docker daemon, then build all six images:

```
minikube docker-env --shell powershell | Invoke-Expression

docker build -t patient-service:latest ./services/patient-service
docker build -t doctor-schedule-service:latest ./services/doctor-schedule-service
docker build -t appointment-service:latest ./services/appointment-service
docker build -t prescription-service:latest ./services/prescription-service
docker build -t billing-service:latest ./services/billing-service
docker build -t demo-ui:latest ./services/demo-ui

minikube docker-env --unset | Invoke-Expression
```

4.4 Create Namespace and Apply Kubernetes Manifests

```
kubectl create namespace hms
kubectl apply -f infra/k8s/hms-storage.yaml -n hms
kubectl apply -f infra/k8s/hms-config.yaml -n hms
kubectl apply -f infra/k8s/hms-secret.yaml -n hms
kubectl apply -f infra/k8s/patient-service.yaml -n hms
kubectl apply -f infra/k8s/doctor-schedule-service.yaml -n hms
kubectl apply -f infra/k8s/appointment-service.yaml -n hms
kubectl apply -f infra/k8s/prescription-service.yaml -n hms
kubectl apply -f infra/k8s/billing-service.yaml -n hms
kubectl apply -f infra/k8s/prometheus-config.yaml -n hms
kubectl apply -f infra/k8s/prometheus-deployment.yaml -n hms
kubectl apply -f infra/k8s/grafana-deployment.yaml -n hms
kubectl apply -f infra/k8s/demo-ui.yaml -n hms
```

4.5 Wait for All Pods to Be Ready

```
kubectl get pods -n hms -w
```

```
C:\Users\HP\Documents\WILP\Sem II\ScalableServices\Assignment\ss-assignment-3>kubectl get pods -n hms -w
NAME                                    READY   STATUS    RESTARTS   AGE
appointment-service-5c99f859b-b97n6     1/1     Running   0          100m
billing-service-76bb7d58d6-98n2p        1/1     Running   0          100m
demo-ui-56ddccdc76-dk8q9                1/1     Running   0          100m
doctor-schedule-service-78c47c4f87-m84p9 1/1    Running   0          100m
grafana-7d7874b9c9-bf2c8                1/1     Running   0          100m
patient-service-8467bb999d-9wkjr        1/1     Running   0          100m
prescription-service-6f5d59ddf5-9qk6p   1/1     Running   0          100m
prometheus-6c5c964b56-p6p4v             1/1     Running   0          100m
```

4.6 Verify Minikube Cluster (Evidence)

```
kubectl get nodes
kubectl config current-context
```

```
C:\Users\HP\Documents\WILP\Sem II\ScalableServices\Assignment\ss-assignment-3>kubectl get nodes
NAME       STATUS   ROLES           AGE     VERSION
minikube   Ready    control-plane   3h40m   v1.35.1

C:\Users\HP\Documents\WILP\Sem II\ScalableServices\Assignment\ss-assignment-3>kubectl config current-context
minikube
```

4.7 Start Port-Forwarding and Seed the Database

Open **five background jobs** for port-forwarding:

```
Start-Job { kubectl port-forward -n hms svc/patient-service 5001:5001 }
Start-Job { kubectl port-forward -n hms svc/doctor-schedule-service 5002:5002 }
Start-Job { kubectl port-forward -n hms svc/appointment-service 5003:5003 }
Start-Job { kubectl port-forward -n hms svc/prescription-service 5004:5004 }
Start-Job { kubectl port-forward -n hms svc/billing-service 5005:5005 }
```

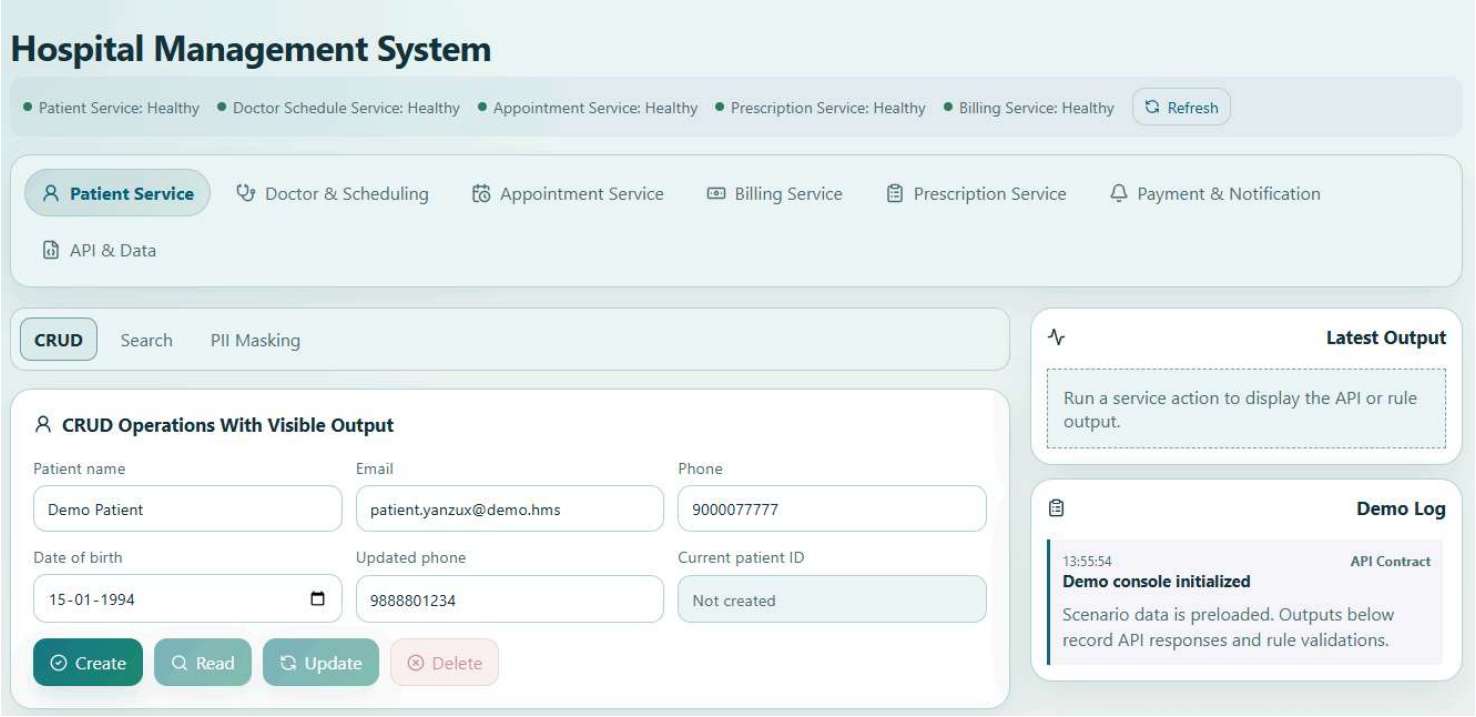
Wait 5 seconds, then seed:

```
cd services
python seed_all.py
```

```
C:\Users\HP\Documents\WILP\Sem II\ScalableServices\Assignment\ss-assignment-3\services>python seed_all.py
Waiting for all services to be healthy...
  OK patient-service ready
  OK doctor-schedule-service ready
  OK appointment-service ready
  OK prescription-service ready
  OK billing-service ready

=== Seeding Patients ===
  Done: 60/60 patients

=== Seeding Doctors ===
  Done: 25/25 doctors

=== Seeding Appointments ===
  Done: 300/300 appointments (148 slot_end values fixed)

=== Seeding Prescriptions ===
  Done: 220/220 prescriptions

=== Seeding Bills ===
  Done: 300/300 bills

=== Seeding Payments ===
  Done: 150/150 payments

=== Seeding complete ===
```

4.8 Access the Demo UI

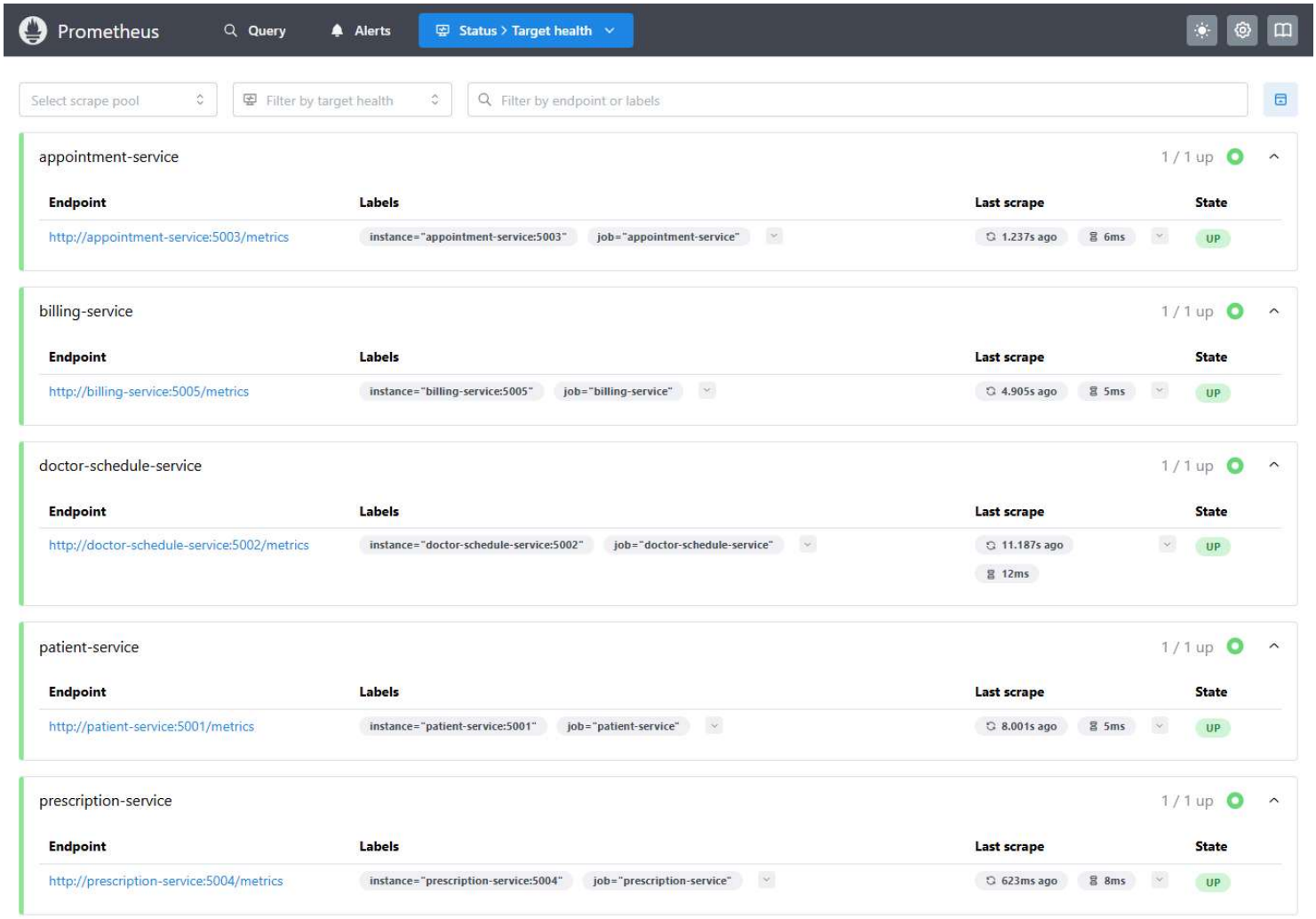
```
minikube service demo-ui -n hms --url
```



4.9 Access Prometheus

Port-forward Prometheus and open it in your browser:

```
Start-Job { kubectl port-forward -n hms svc/prometheus 9090:9090 }
```



4.10 Access Grafana

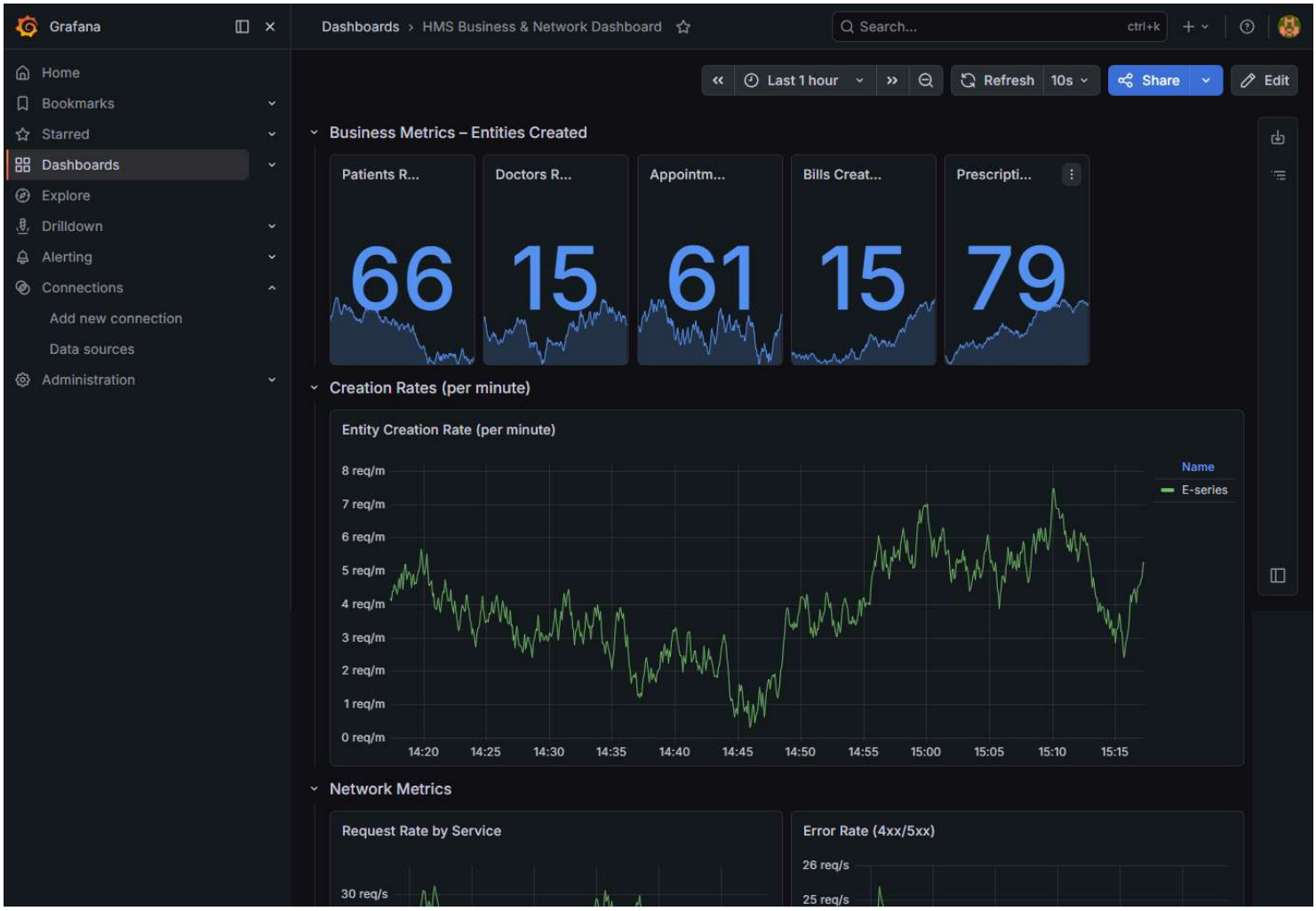
Port-forward Grafana and open it in your browser:

```
Start-Job { kubectl port-forward -n hms svc/grafana 3000:3000 }
```

Navigate to `http://localhost:3000`. Login with default credentials (`admin` / `admin`).

Add Prometheus as a data source:

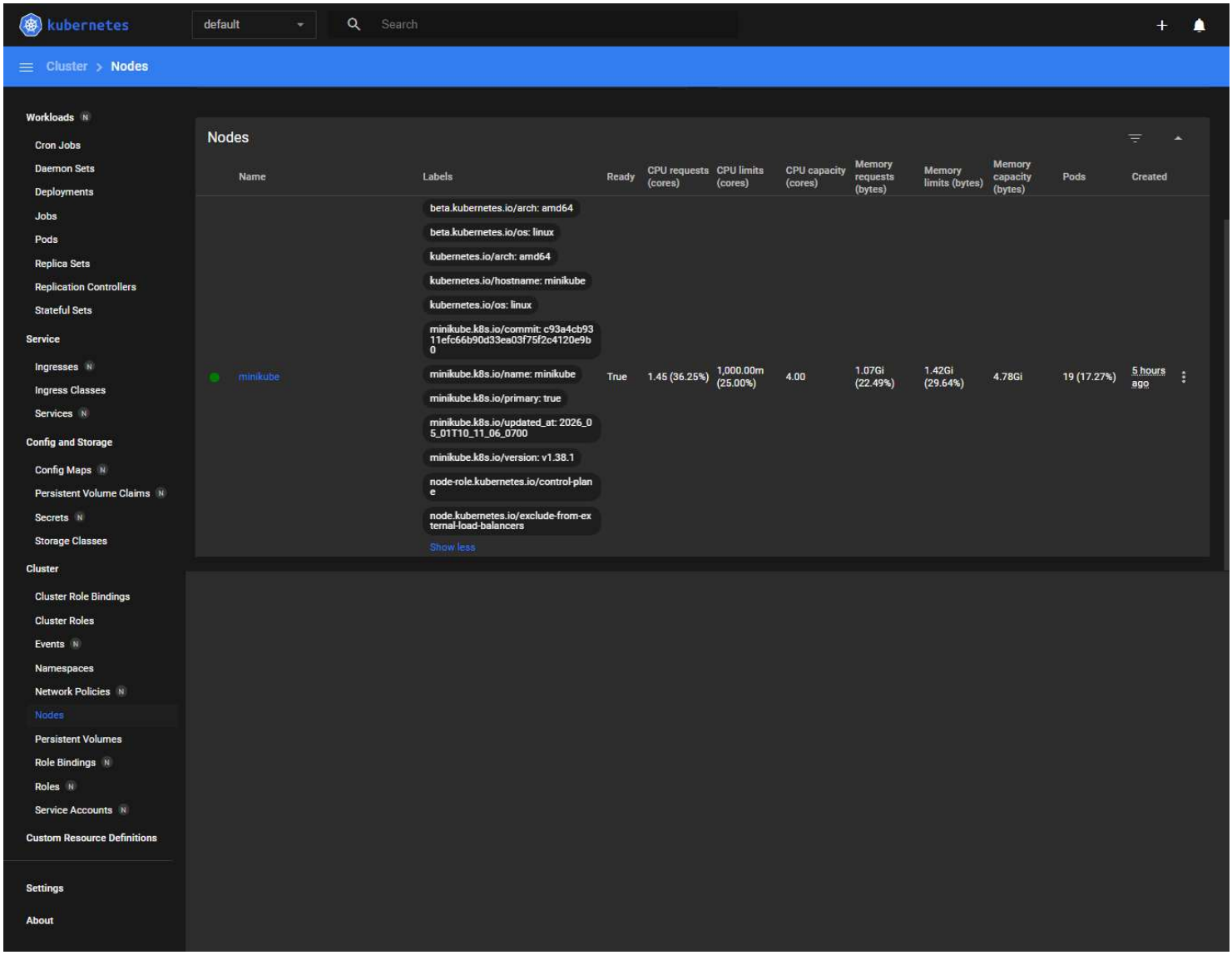
1. Go to **Configuration** → **Data Sources** → **Add data source**.
2. Select **Prometheus**.
3. Set URL to `http://prometheus:9090`.
4. Click **Save & Test**.



4.11 Explore Kubernetes Features (Optional Demos)

Minikube Dashboard:

```
minikube dashboard
```



Self-Healing Demonstration:

```
kubectl delete pod -n hms -l app=patient-service
kubectl get pods -n hms -w
```

Observe Kubernetes automatically recreating the terminated pod.

View Pod Logs:

```
kubectl logs -n hms deployment/patient-service --tail=20
```

Check Resource Usage:

```
minikube addons enable metrics-server
kubectl top pods -n hms
```

4.12 (Alternative) One-Click Deployment with restart-hms.ps1

For a complete clean restart, use the automated PowerShell script:

```
.\restart-hms.ps1
```

This script performs all steps 4.1–4.7 automatically:

- Deletes existing `hms` namespace
- Rebuilds all Docker images inside Minikube
- Recreates namespace and applies all manifests
- Creates the `doctor-service` alias
- Waits for all pods to be ready
- Starts port-forwarding background jobs
- Runs the seed script
- Prints the Demo UI URL

4.13 Stop and Clean Up

```
# Stop port-forwarding jobs
Get-Job | Stop-Job
Get-Job | Remove-Job

# Delete the namespace (removes all resources)
kubectl delete namespace hms

# Stop Minikube
minikube stop
```

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5. GitHub Repository Links

Service	Repository URL
Consolidated Repository	https://github.com/adi-bits-2025/ss-assignment-3
Patient Service	https://github.com/adi-bits-2025/patient-service
Doctor Schedule Service	https://github.com/adi-bits-2025/doctoer-schedule-service
Appointment Service	https://github.com/adi-bits-2025/appointment-service
Prescription Service	https://github.com/adi-bits-2025/prescription-service
Billing Service	https://github.com/adi-bits-2025/billing-service

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